Name: ______________________________ Date: ____________________

Physics 438A Group Activity #7

1. Given $[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k$ and $J^2 = J_x^2 + J_y^2 + J_z^2$, show that $[J^2, J_i] = 0$ for all components of $\mathbf{J}$. Briefly explain why it's enough to show that it works for just one component.

2. Given $J_+ = J_x + iJ_y$ and $J_- = J_x - iJ_y$, show that $[J_z, J_\pm] = \pm\hbar J_\pm$ and $[J_+, J_-] = 2\hbar J_z$.